

Homework 3

Nervous and Bone mechanics

$$1. \frac{dv}{dt} = f(v) - w + I_a = 0 \Rightarrow w = f(v) + I_a$$

$$\frac{dw}{dt} = bv - w = 0 \Rightarrow w = bv$$

if the model have a periodic behaviour

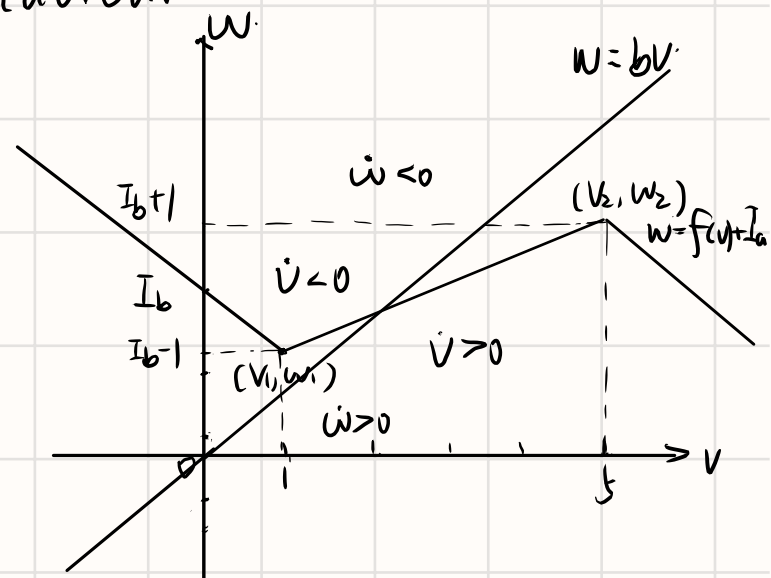
I_a and b should be:

$$I_1 < I_a < I_2, \quad b > \tan \theta = \frac{w_2 - w_1}{v_2 - v_1}$$

$$\text{so, } I_a > I_1 = bv_1 - f(v_1) = b + 1$$

$$I_a < I_2 = bv_2 - f(v_2) = 5b - 1$$

$$b > \frac{w_2 - w_1}{v_2 - v_1} = \frac{I_a + 1 - (I_a - 1)}{5 - 1} = \frac{2}{4} = \frac{1}{2}$$



In conclusion, $b > \frac{1}{2}$, $I_a \in (b+1, 5b-1)$

$$2. I_A = \int r^2 dm$$

$$\text{for A, } I_A = 4 \int_{-1}^1 y^2 \cdot 2 dy + 2 \int_{-5}^5 y^2 \cdot 2 dy = 8 \frac{y^3}{3} \Big|_{-1}^1 + 4 \frac{y^3}{3} \Big|_{-5}^5 = \frac{1016}{3} \text{ cm}^4$$

$$\begin{aligned} \text{for B, } I_B &= \int_{-6}^6 y^2 \cdot 2 dy + \left(\int_{-4}^4 y^2 \cdot 8 dy - \int_{-2}^2 y^2 \cdot 8 dy \right) \\ &= 2 \frac{y^3}{3} \Big|_{-6}^6 + 8 \left(\frac{y^3}{3} \Big|_{-4}^4 - \frac{y^3}{3} \Big|_{-2}^2 \right) = \frac{1760}{3} \text{ cm}^4 \end{aligned}$$

so, $I_B > I_A$.

since $\frac{1}{R} = \frac{M_b}{EI_A}$, B shape has a bigger moment of inertia, will resist the bending better.